Techno-Economic Assessment of Steel Slag (BOF) Valorization

H_2 Production, CO_2 Sequestration, and Iron Recovery

Separation-Agents Automated Valorization Pipeline
`sep_agents v0.3`

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Abstract

This report presents a techno-economic analysis of valorizing Basic Oxygen Furnace (BOF) steel slag at a throughput of 500 t/d. Six value streams were identified: iron concentrate recovery, CO_2 mineral carbonation, H_2 production via serpentinization, manganese recovery, vanadium/chromium extraction, and construction aggregate. Four process topologies were evaluated using a Generalized Disjunctive Programming (GDP) superstructure framework, with economics computed using a JAX-based techno-economic model. The optimal topology (Config-4: minimal process without auxiliary heating or scrubbing) yields a net cost of −6.83 USD/t slag at current commodity prices. Breakeven is achieved at a CO_2 credit price of ≥90 USD/t or acid consumption ≤100 kg/t. Under an optimistic scenario (CO_2 = 100 USD/t, acid = 100 kg/t), the project generates 9.85 USD/t net profit (1.63 MUSD/yr). The analysis was performed using the Reaktoro thermodynamic engine, IDAES-PSE flowsheet solver, and JAX automatic differentiation for cost sensitivity analysis.

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1 Introduction and Opportunity

Steel slag is the primary by-product of steelmaking, generated at rates of 100–200 kg per tonne of crude steel. Globally, over 200 Mt/yr of BOF slag is produced, with significant fractions landfilled due to limited utilization pathways. BOF slag is rich in CaO (35–45 %), FeO (15–25 %), SiO_2 (10–20 %), and MgO (5–10 %), with minor quantities of MnO, Cr_2O_3, V_2O_5, P_2O_5, and TiO_2.

This mineral composition creates multiple simultaneous valorization opportunities:

1. **CO_2 mineralization:** High CaO and MgO content provides alkalinity for mineral carbonation, permanently sequestering CO_2 as $CaCO_3$/$MgCO_3$ and generating carbon credits.
2. **H_2 production:** Fe^{2+}-bearing silicates can produce hydrogen via serpentinization-like reactions at elevated temperature and pressure.
3. **Iron recovery:** Magnetite and hematite fractions can be concentrated via magnetic separation for direct sale.
4. **Critical mineral recovery:** Vanadium and chromium represent high-value co-products.

The objective of this study is to determine the optimal process configuration that maximizes net economic value from a 500 t/d BOF slag feedstock.

2 Raw Material Composition

2.1 Bulk Oxide Chemistry

Table 1: BOF Steel Slag Bulk Oxide Composition

Oxide	**wt%**	**Role in Valorization**
CaO	40.0	CO_2 carbonation feedstock
FeO	20.0	Iron recovery, H_2 production
SiO_2	15.0	Aggregate, silica co-product
MgO	8.0	CO_2 carbonation, serpentinization
Al_2O_3	5.0	Aggregate, cement additive
MnO	5.0	Manganese recovery
Cr_2O_3	2.0	Ferrochrome production
P_2O_5	1.5	Phosphate fertilizer
V_2O_5	0.5	Ferrovanadium production
TiO_2	0.5	Titanium dioxide pigment
Other / LOI	2.5	—

2.2 Equilibrium Speciation (Acid Leach)

The slag was computationally leached in 2 M HCl at 25 °C using the Reaktoro thermodynamic engine (SUPCRT database, Pitzer activity model).

Table 2: Speciation Results — HCl Acid Leach at 25 °C

Parameter	**Value**	**Unit**
pH	1.09	—
Eh	0.14	V
Ionic Strength	2.14	mol/kg
Species	**Concentration**	**mol/kg**
Cl^-	1.238	
Ca^{2+}	0.252	
$CaCl^+$	0.160	
Fe^{2+}	0.135	
$FeCl^+$	0.115	
$SiO_2(aq)$	0.094	
$H_4SiO_4(aq)$	0.086	
Mg^{2+}	0.073	
Al^{3+}	0.070	
Mn^{2+}	0.026	

3 Process Description

The process consists of two primary reactors (serpentinization and carbonation), supported by heat exchangers, pumps, separators, and optional scrubbing/heating units.

3.1 Process Flow Diagram

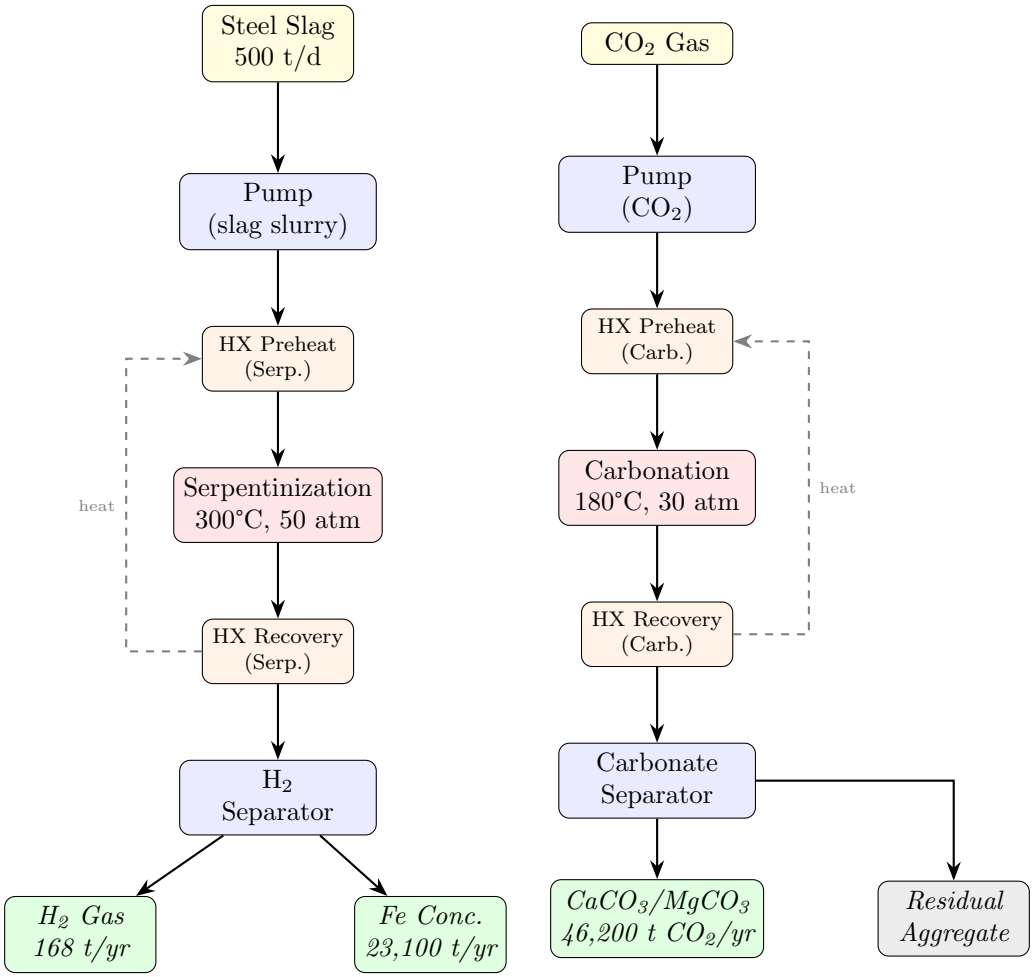

Figure 1: Process flow diagram for Config-4 (minimal topology): steel slag to H_2, $CaCO_3$, and Fe concentrate.

3.2 Key Reactions

Serpentinization (H_2 production):

$$3Fe_2SiO_4 + 2H_2O \longrightarrow 2Fe_3O_4 + 3SiO_2 + 2H_2 \uparrow \tag{1}$$

Mineral Carbonation (CO_2 sequestration):

$$CaO + CO_2 \longrightarrow CaCO_3 \qquad MgO + CO_2 \longrightarrow MgCO_3 \tag{2}$$

4 Modeling Approach

Table 3: Three-Tier Modeling Hierarchy

Tier	Tool	Application
Thermodynamic	Reaktoro GEM (SUPCRT, Pitzer)	Equilibrium speciation, pH, activity coefficients
Process Simulation	IDAES-PSE / Pyomo	Flowsheet mass & energy balances, GDP superstructure optimization
Economic	JAX TEA + autodiff	Itemized CAPEX/OPEX, EAC, cost sensitivity via ∂EAC/∂param

5 GDP Superstructure Analysis

5.1 Superstructure Definition

The `steel_slag_h2_co2` superstructure consists of 15 unit operations with one disjunction (auxiliary heater vs. full heat recovery) and one optional unit (inter-reactor scrubber). This produces $2 \times 2 = 4$ feasible topologies.

5.2 Topology Ranking

Table 4: Ranked Topology Comparison (500 t/d, 330 operating days/yr)

Rank	Topology	EAC (M$/yr)	Revenue (M$/yr)	Net (M$/yr)	Net/t ($/t)
1	Minimal (no heater, no scrub)	6.73	5.60	−1.13	−6.83
2	Heat recovery + scrubber	7.47	5.84	−1.62	−9.85
3	Aux heater, no scrubber	8.96	5.72	−3.24	−19.65
4	Full (heater + scrubber)	9.95	5.94	−4.01	−24.33

Key finding: The minimal topology (Config-4) is optimal because the marginal recovery improvement from auxiliary heating and scrubbing does not offset the additional energy and reagent costs.

6 Cost Model Assumptions

Table 5: Plant-Level Parameters

Parameter	**Value**	**Unit**
Slag throughput	500	t/d
Operating days	330	d/yr
Discount rate	8	%
Project lifetime	20	yr
Bond work index (slag)	12	kWh/t
Acid consumption	150	kg HCl / t slag
Serpentinization T	300	°C
Carbonation T	180	°C

Table 6: Commodity Prices

Product	**Price**	**Source**
H_2 (green)	\$4,000/t	DOE target
CO_2 credit	\$60/t CO_2	Voluntary market
Fe concentrate	\$120/t	FOB
Mn (electrolytic)	\$2,500/t	LME
V_2O_5	\$12,000/t	Asian Metal

7 Results

7.1 Itemized Economics (Config-4)

Table 7: Itemized CAPEX and OPEX — Config-4 (Minimal)

Cost Item	**Annual Cost**	**\$/t slag**
CAPEX (annualized)		
Comminution	\$624,000	3.78
Leaching	\$400,000	2.42
Processing	\$150,000	0.91
OPEX		
Comminution (energy)	\$164,000	0.99
Acid (HCl)	\$4,950,000	30.00
Other processing	\$251,000	1.52
Total EAC	**\$6,730,000**	**40.79**

Table 8: Revenue Breakdown — Config-4

Product	**Production**	**Price**	**Revenue**	**\$/t slag**
H_2 gas	168 t/yr	\$4,000/t	\$673,000	4.08
CO_2 credits	37,884 t/yr	\$60/t	\$2,273,000	13.78
Fe concentrate	23,100 t/yr	\$120/t	\$2,772,000	16.80
Total			**\$5,600,000**	**33.96**

7.2 Net Economics

$$\text{Net Value} = \$5.60\text{M/yr} - \$6.73\text{M/yr} = -\$1.13\text{M/yr} \quad (-\$6.83/\text{t slag}) \tag{3}$$

7.3 Sensitivity Analysis

Table 9: CO_2 Credit Price Sensitivity (Config-4, acid = 150 kg/t)

CO_2 Price (\$/t)	**Net Value (M\$/yr)**	**Net/t (\$/t)**
40	−1.89	−11.43
60	−1.13	−6.83
yellow!20 80	−0.37	−2.24
green!10 100	+0.39	+2.35
120	+1.15	+6.94
150	+2.28	+13.83
200	+4.18	+25.31

Table 10: Acid Consumption Sensitivity (Config-4, CO_2 = \$60/t)

Acid (kg/t)	**Net Value (M\$/yr)**	**Net/t (\$/t)**
50	+1.35	+8.17
75	+0.73	+4.42
green!10 100	+0.11	+0.67
125	−0.51	−3.08
150	−1.13	−6.83
200	−2.37	−14.33

7.4 Top Cost Drivers (JAX Autodiff)

Table 11: Cost Sensitivity via JAX Automatic Differentiation

Parameter	**∂EAC/∂param**	**Unit**
Acid consumption	+24,750	\$/yr per kg/t
Throughput	+16,002	\$/yr per t/d
Bond work index	+15,840	\$/yr per kWh/t
Residence time	+10,185	\$/yr per hr
Operating temperature	+8,250	\$/yr per °C

8 Discussion

8.1 Economic Drivers

The dominant cost is HCl acid consumption at \$30/t slag, representing 74 % of total OPEX. This suggests that acid-free or acid-recycling process routes should be prioritized in future work. Revenue is dominated by iron concentrate (49 %) and CO_2 credits (41 %), with H_2 contributing only 10 % due to relatively low yield.

8.2 Breakeven Conditions

Two pathways to economic viability exist:

1. **Higher CO_2 prices**: Breakeven at $\geq$90 USD/t, which is within the range of compliance-grade carbon markets (EU ETS: 60 EUR/t to 100 EUR/t in 2024–2026).
2. **Reduced acid consumption**: Breakeven at $\leq$100 kg/t, achievable via slag pre-treatment, selective dissolution, or acid regeneration loops.

Under a combined optimistic scenario (CO_2 = \$100/t, acid = 100 kg/t), the project generates 9.85 USD/t profit (1.63 MUSD/yr).

8.3 Key Risks

- **CO_2 credit price volatility**: Revenue is highly sensitive to carbon market conditions.
- **Slag variability**: BOF slag composition varies by steel grade and furnace operation.
- **Scale-up risk**: Serpentinization kinetics at 300 °C are not fully validated at industrial scale.
- **Cr/V not modeled**: These potentially high-value co-products are excluded from the Reaktoro speciation model and revenue estimates.

9 Conclusions

1. BOF steel slag at 500 t/d can produce 168 t/yr H_2, sequester 37,884 t/yr CO_2, and recover 23,100 t/yr iron concentrate.

2. The optimal topology (Config-4: minimal process) has an EAC of 6.73 MUSD/yr against revenue of 5.60 MUSD/yr, yielding a net cost of −6.83 USD/t.

3. **Acid consumption is the dominant cost driver**, representing 74 % of OPEX. Process improvements targeting acid reduction or recycling are the highest-priority R&D targets.

4. Breakeven is achievable under reasonable near-term scenarios: CO_2 credits $\geq$90 USD/t (compliance market) or acid consumption $\leq$100 kgHCl/t (acid regeneration).

5. Under optimistic conditions, the project generates 1.63 MUSD/yr (9.85 USD/t).

6. **Recommended next steps**: pilot-scale serpentinization trials, acid regeneration feasibility study, and inclusion of V/Cr co-product economics.

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